

B-221 Seat No.
B. E. Sem. III (IT) (Reg.) Examination
December - 2002
Data Structure & Algorithm

Time : 3 Hours]

[Total Marks : 100

- Instructions :** (1) Answer to the two sections must be written in separate answer sheet.
(2) Figures to the right indicate full marks.
(3) Assume suitable data, if necessary.

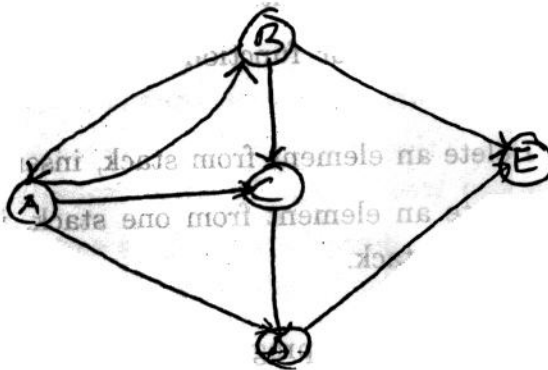
SECTION - I

- 1 Answer the following : 18
- (a) Explain with examples, Primitive Data structure and non-primitive Data structure. 3
- (b) Explain recursion and give algorithm for "Tower of Hanoi" problem with N Discs. Is recursion use stack? 5
- (c) What is Queue ? Write algorithm or function for following : 5
- (i) Insert an element
- (ii) Delete an element.
- (d) Write Algorithm or function for following situations using array : 5
- (i) Delete an element from stack, insert it in queue.
- (ii) Delete an element from one stack, insert it into second stack.
- 2 (a) Write algorithm or program to implement following in single linked list : 8
- (i) Insert an element at any place
- Reverse the linked list.

- (b) What is circular queue ? Implement following in circular queue :
- Insert an element
 - Delete an element

OR

- 2 (a) What is double linked list ? Implement following in double linked list : 10
- Sort the value in list
 - Reverse the list.
- (b) What is infix, post fix, prefix notation ? Translate infix $(A + B \uparrow C \uparrow D) * (E + F / D)$ to post fix notation. Calculate value for $A = 10, B = C = D = 2, E = D = 1, F = 2$. 6
- 3 Answer the following : (any four) 16
- What is simulation ? Explain with example.
 - What is hashing ? Where it is used ? Explain with example.
 - Give the "Row-major" and "Column-major" representation for $n \times m$ direction matrix (array).
 - Give the adjacency list, storage representation for adjacency list and edge list of the following graph :



- (e) Give the application of graph.

SECTION - II

- 4 Answer the following : 18
- (a) What is searching ? Give the algorithm for Binary search :
 - (b) Differentiate : BFS and DFS.
 - (c) Explain quicksort on following data :
41, 22, 74, 11, 64, 57, 93, 35, 98, 86
- 5 (a) What is time and space complexity in any sorting algorithm ? Give the algorithm for insertion sort with any suitable data : (at least 7) 8
- (b) Give an algorithm for insert a node in binary tree. 8
- OR
- 5 (a) Give the algorithm for Heap Sort. 8
- (b) Give an algorithm for delete a node in binary tree. 8
- 6 Answer the following : (any four) 16
- (a) Define following terms :
 - (i) Cycle
 - (ii) Weighted graph
 - (iii) In-degree
 - (iv) Ordered tree.
 - (b) Explain indexed file organization.
 - (c) Explain pre-order tree traversal and give algorithm.
 - (d) What is the difference between binary tree and threaded binary tree ? Explain with example.
 - (e) Give the algorithm for bubble sort and also give time complexity of bubble sort in Average case analysis.
 - (f) Explain various multi key access file organization.